

Covariant Quantum Memory and Emergent Goal Geometry

Exact Qutrit Torus Metrics, Predictive Fibers, and Opaque Action Learning

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Abstract

We ask whether a three-dimensional quantum system can support an operationally learned two-dimensional space when action meanings are inferred from future observations rather than supplied as coordinate labels. The construction uses Weyl-covariant qutrit instruments on the nine Hesse symmetric informationally complete states. A correction of terminal semantics is decisive. If success means that the current predictive state has reached the goal, the original integrated Hesse instrument has exact Bellman shells

$$V_{\text{self}} = 0, \quad V_{\text{edge}} = 4, \quad V_{\text{diagonal}} = 5.$$

This is a nondegenerate translation-invariant metric on $\mathbb{Z}_3^2$. The earlier shells 4, 4, 5 answer the different report-the-goal-again task and are nonmetric. We prove a general theorem under which positive covariant hitting costs form a group metric and derive the exact Schoenberg spectrum of the Hesse metric: it is Euclidean of minimal rank eight, so it recovers a two-dimensional torus topology but not a global planar embedding.

We introduce an exactly soluble integrated family with an observed unitary memory branch and informative Hesse reset branches. For sharp reports its Bellman shells are

$$E = \frac{4}{(1 + \mu)^2}, \quad D = \frac{3\mu + 5}{(1 + \mu)^2}.$$

It continuously interpolates between informative reset geometry and the exact torus word metric. The analytic value $\mu = (4\sqrt{2} - 5)/3$ makes every elementary cell an exact Euclidean square. A deterministic 380-point search selects $(\mu, \xi) = (0.8, 1)$ as a high-memory compromise: 0.050326 bits of immediate information, shells $(0, 1.234568, 2.283951)$, and 3.33% scaled distortion from torus word distance. An observable common-probe protocol identifies this exact family's action maps using only opaque anchor/action/future-report triples and recovers the order-nine commuting translation group. A separate opaque Lüders experiment recovers the same token topology from unsegmented histories.

For the outcome-discarded channel, equal-length paths with the same displacement close exactly, while cancelling detours accumulate different noise. That noise age is not an extra state for an agent observing every branch: its conditional state remains a Hesse ray. Separate Lüders-instrument experiments confirm latent future-predictive memory while showing that naive finite suffix models do not learn it reliably. We therefore find a candidate fiber, not an already learned bundle. We state reproducible acceptance criteria for larger tori, local flat patches, and spectral predictive-state learning.

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1 Motivation: geometry from attainable goals

Quantum hodology asks whether spatial distance can be reconstructed from an agent’s goal-directed behavior. A goal reached cheaply and reliably is near; one requiring long, risky, or costly intervention is far. If optimal goal costs organize themselves into a low-dimensional homogeneous geometry, an agent’s controlled trajectory may be interpreted as motion in emergent space.

This requires more than plotting assigned state labels. A convincing construction must answer three separate questions:

1. Can the agent distinguish predictive situations using future statistics?
2. Can it learn what opaque actions do without coordinate names?
3. Do optimal goal costs satisfy the axioms and local structure of the claimed geometry?

A classical counter can trace a grid even when quantum observations contain no corresponding structure. Conversely, quantum actions can be distinguishable without forming translations.

The present study retains quantum state across selected observed branches. This suggests a future base/fiber geometry: the base records place-like translation on a finite torus, while a fiber could record purity, posterior phase, or hidden memory. In the exact observed-branch family, nonselective noise age appears only when outcomes are discarded. It also forces an honest distinction between global torus topology and a locally flat planar patch.

Contributions.

- First-principles controlled-instrument and predictive semantics.
- An exact terminal-semantics correction and proof of Hesse shells $(0, 4, 5)$.
- A general covariant hitting-cost metric theorem.
- Exact Fourier–Schoenberg classification of the global metric.

- Two retained-memory constructions: an exactly finite observed memory/reset family and a Lüders-like latent-memory family.
- Closed-form Bellman values, an analytic local-square point, numerical search, opaque learning, controls, and a larger-torus program.

2 Concrete operational specification: actions, starts, and goals

This section states the experimental interface before developing its theory. There are three related protocols in the paper, and they intentionally use different initialization procedures. The exact Bellman experiment defines the goal geometry; the common-probe experiment learns the meanings of its opaque actions; and the Lüders experiment asks whether longer histories carry predictive memory. They should not be conflated.

2.1 The qutrit and its nine candidate places

In every quantum protocol the physical Hilbert space is $\mathcal{H} = \mathbb{C}^3$. The nine candidate place states are the Hesse SIC projectors

$$\Pi_{mn} = |\psi_{mn}\rangle\langle\psi_{mn}|, \quad (m, n) \in \mathbb{Z}_3^2,$$

defined explicitly in [Section 5](#). They are nonorthogonal: different places have overlap $\text{Tr}(\Pi_s \Pi_t) = 1/4$. The coordinate pair (m, n) is used by the theorist and by offline scoring only. The agent receives persistent but otherwise arbitrary integer tokens.

2.2 Exact state-hitting navigation experiment

The navigation action alphabet contains exactly five choices,

$$\mathcal{A} = \{a_0, a_{+x}, a_{-x}, a_{+y}, a_{-y}\}, \quad U_a \in \{I, X, X^\dagger, Z, Z^\dagger\}. \quad (1)$$

The symbols $\pm x, \pm y$ are privileged explanatory names. In the opaque experiment the agent sees only five shuffled button IDs. Pressing a button does *not* let the agent choose a Kraus branch. It invokes the complete instrument associated with that button, Nature samples an observed outcome, and the agent sees that outcome.

For the principal memory/reset family, button a has ten possible observed outcomes:

$$\mathcal{O} = \{r, q_0, q_1, \dots, q_8\}.$$

Outcome r occurs with probability μ and applies the full-rank memory map $\rho \mapsto U_a \rho U_a^\dagger$. Outcome q_o applies the reset branch of [Eq. \(12\)](#), reports token o , and prepares Π_o . For a current Hesse class s , the operational update rule is therefore

$$s \xrightarrow{a, r} s + a, \quad s \xrightarrow{a, q_o} o. \quad (2)$$

Each button press costs one unit, irrespective of its outcome. There is no separate free move, free measurement, or hidden coordinate update.

The initial state for an individual navigation episode is one known Hesse state,

$$\rho_0 = \Pi_{s_0}, \quad s_0 \in \mathbb{Z}_3^2. \quad (3)$$

One may take $s_0 = (0, 0)$ as a canonical example. The reported distance matrix does not privilege that choice: it solves all 9×9 ordered start–goal problems, so s_0 ranges over all nine projectors. Translation covariance makes different origins equivalent. Operationally, a known start can be established by the anchor preparation described below; no coordinate pair is then revealed to the agent.

There are exactly nine place goals, one for each predictive class:

$$G_g = \{\text{the current predictive state is } \Pi_g\}, \quad g \in \mathbb{Z}_3^2. \quad (4)$$

Goal G_g is completed at the first hitting time

$$\tau_g = \inf\{t \geq 0 : S_t = g\}.$$

The objective is to minimize $\mathbb{E}[\tau_g \mid S_0 = s_0]$. Consequently an episode initialized at its goal ends immediately and costs zero: $V_g(g) = 0$. The goal is *not* a requested action sequence and is not the event of seeing one more report token. The agent is supplied the desired opaque goal token and may condition its policy on that token.

At $\mu = \mu_\square$, the phrase “exact Euclidean square” refers to the four-goal metric subspace

$$\{G_{00}, G_{10}, G_{01}, G_{11}\}.$$

It does not add diagonal buttons or change the goals. It says that the optimal costs among these four existing goals have edge/diagonal ratio $1 : \sqrt{2}$. The full collection of nine goals retains toroidal identifications.

2.3 The report-again comparison is a different goal

For the baseline integrated Hesse instrument of Eq. (9), the available buttons are still precisely the five in Eq. (1), but each press has nine observed outcomes $o \in \{0, \dots, 8\}$ and prepares Π_o . Its navigation start is again Π_{s_0} .

The comparison task called “report again” does not terminate merely because $S_t = g$. Its terminal event is

$$G_g^{\text{report}} = \{\text{after another chosen button press, the observed outcome token is } g\}. \quad (5)$$

Thus an agent beginning in Π_g must still pay for at least one intervention. This event goal gives shells $(4, 4, 5)$, whereas the place goal in Eq. (4) gives $(0, 4, 5)$. Nothing about the instrument or initial density operator changes; only the success event changes.

2.4 Observable action-calibration protocol

The common-probe study is a calibration experiment, not an extra navigation policy. In each trial:

1. A common sharp Hesse anchor instrument produces an opaque token s and thereby prepares Π_s . Across the dataset, all nine anchors and all five buttons are sampled equally often.
2. The agent presses one of the same five opaque navigation buttons in Eq. (1). The trial is retained when the publicly observed branch token is r ; no hidden branch selection is given to the learner.
3. The same sharp Hesse instrument, with identity control, is applied as a common future probe and emits an opaque report token o .

The learning record is only (s, a, o) . It contains neither a density matrix nor a translated-state label. The physical initial state is freshly prepared as Π_s on each trial rather than fixed to one projector. Hidden coordinates enter only after learning, to score the inferred permutations up to a persistent token gauge.

Operationally, the balanced anchor design can be implemented by beginning anchor trials at $I/3$, for which the sharp Hesse outcomes are uniform, and collecting an equal number of trials in each observed anchor/button stratum. The simulation directly conditions on those nine observed anchor strata; it does not expose their hidden (m, n) coordinates to the learner.

2.5 Opaque Lüders sequence-learning protocol

The independent history-learning experiment uses the same five unitary parts $I, X, X^\dagger, Z, Z^\dagger$, but each button invokes the nine-outcome Lüders instrument $K_{a,o}^\eta = \sqrt{E_o^\eta} U_a$ of [Section 8.5](#). The agent observes one of nine shuffled outcome tokens; there is no separately observed memory token. Every seven-step sequence is initialized at

$$\rho_0 = I/3. \quad (6)$$

During data collection, opaque buttons are sampled uniformly in order to learn controlled future laws. No goal reward is used to fit the predictive model. After fitting, each of the nine learned outcome/predictive tokens is in turn declared a state-hitting goal, with zero terminal value at that token, and the learned token kernel is used to solve nine goal-conditioned Bellman problems. For $0 < \eta < 1$ this token-level planner is approximate because the last token is not the full predictive state. The density-matrix filter is used only as an offline prediction oracle.

Protocol summary. In exact navigation, the agent chooses among five memory/reset instruments, starts in a known Π_{s_0} , and tries to hit one of nine place goals. In common-probe calibration, an anchor prepares a fresh Π_s , one of those same five buttons is tested, and a common future probe supplies the response; there is no navigation reward. In Lüders history learning, every string starts at $I/3$, the agent samples five nine-outcome Lüders instruments, and the immediate task is held-out prediction; nine approximate token-hitting goals are solved only after learning. In all three cases, “five” means the identity and four cardinal Weyl controls, presented under opaque IDs whenever action meaning is being learned.

3 Controlled quantum processes

3.1 States, channels, and instruments

Let $\mathcal{H} \simeq \mathbb{C}^3$. A density operator satisfies $\rho \succeq 0$, $\text{Tr } \rho = 1$. For action $a \in \mathcal{A}$, an instrument is

$$\mathfrak{I}^a = \{\mathcal{E}_o^a : o \in \mathcal{O}\}$$

where each branch is completely positive and trace nonincreasing and $\Phi_a = \sum_o \mathcal{E}_o^a$ is trace preserving [1, 2]. A Kraus representation is

$$\mathcal{E}_o^a(\rho) = \sum_{\alpha} K_{a,o,\alpha} \rho K_{a,o,\alpha}^\dagger, \quad \sum_{o,\alpha} K_{a,o,\alpha}^\dagger K_{a,o,\alpha} = I.$$

The outcome and posterior are

$$p(o \mid \rho, a) = \text{Tr } \mathcal{E}_o^a(\rho), \quad \rho_{a,o} = \frac{\mathcal{E}_o^a(\rho)}{p(o \mid \rho, a)}.$$

Several ranks are distinct. Hilbert dimension counts amplitudes. Branch Choi rank counts minimal Kraus operators. Matrix rank of a Kraus operator controls invertibility on state support. Predictive Hankel rank measures future-statistical dimension. Schoenberg rank measures Euclidean embedding dimension.

3.2 Histories, tests, and predictive equivalence

A history $h = (a_1, o_1) \cdots (a_t, o_t)$ has unnormalized state

$$\tilde{\rho}_h = \mathcal{E}_{o_t}^{a_t} \circ \cdots \circ \mathcal{E}_{o_1}^{a_1}(\rho_0).$$

A future test $\tau = (b_1, y_1) \cdots (b_k, y_k)$ has probability

$$p(\tau \mid h) = \frac{\text{Tr}[\mathcal{E}_{y_k}^{b_k} \circ \cdots \circ \mathcal{E}_{y_1}^{b_1}(\tilde{\rho}_h)]}{\text{Tr} \tilde{\rho}_h}.$$

The controlled Hankel matrix $H_{h,\tau} = p(\tau \mid h)$ contains all operational predictions.

Definition 3.1 (Predictive equivalence). Histories h, h' satisfy $h \sim h'$ when

$$p(\tau \mid h) = p(\tau \mid h')$$

for every admissible future test. Their classes are causal or predictive states.

A predictive-state representation stores probabilities of core tests instead of a privileged density matrix. When the Hankel rank is r , r independent core predictions suffice [4]. A qutrit density operator occupies an eight-dimensional affine space, but restricted controls may lower predictive rank, while a classical memory register may raise total history dimension.

3.3 Action meaning and gauge

An action's operational meaning is its transformation of future probabilities. Actions a, b are distinguishable when some history, current outcome, and continuation test satisfy

$$p(o, \tau \mid h, a) \neq p(o, \tau \mid h, b).$$

Immediate mutual information may vanish although a future probe separates the post-action states.

PSR coordinates admit invertible similarity gauges. Tokens may be permuted; generators of $\mathbb{Z}_m^2$ may be exchanged, inverted, or transformed by a group automorphism. Gate-set tomography obeys the same principle [6, 7]. We therefore evaluate group order, commutation, orbit size, and held-out likelihood before privileged alignment.

4 Covariance and predictive closure

4.1 Covariant instruments

Let $G = \mathbb{Z}_m \times \mathbb{Z}_m$, and let $g \mapsto U_g$ be a possibly projective unitary representation. A base instrument is covariant when

$$\mathcal{J}_{o+g}(U_g \rho U_g^\dagger) = U_g \mathcal{J}_o(\rho) U_g^\dagger. \quad (3.1)$$

Integrated actions take the form

$$\mathcal{E}_o^a = \mathcal{J}_o \circ \text{Ad}_{U_a}. \quad (3.2)$$

Covariance gives

$$p(o + g \mid U_g \rho U_g^\dagger, a) = p(o \mid \rho, a),$$

so absolute labels can be shuffled while relative responses survive.

Proposition 4.1 (Positive Choi-seed parametrization). *Let $R_g = U_g \otimes \overline{U_g}$ act on output–input Choi space. Choose $C_0 \succeq 0$, and put $C_o = R_o C_0 R_o^\dagger$. If*

$$\text{Tr}_{\text{out}} \sum_o C_o = I,$$

the corresponding maps form a CPTP covariant instrument. Conversely, every transitive covariant instrument has this orbit form, with C_0 invariant under the reference outcome’s stabilizer.

Proof. Unitary conjugation preserves Choi positivity. The partial-trace condition is trace preservation of the sum. Conjugation by R_g maps C_o to C_{o+g} , proving covariance. Conversely, covariance transports a reference Choi matrix around its orbit; independence of orbit representative is stabilizer invariance. $\square$

Writing $C_0 = LL^\dagger$ is a practical optimization parametrization: complete positivity and covariance remain exact, while normalization is a linear constraint.

4.2 Nonselective path closure and a prospective fiber

Let $\Phi = \sum_o \mathcal{J}_o$. Covariance implies

$$\Phi \text{Ad}_{U_g} = \text{Ad}_{U_g} \Phi.$$

For $\Phi_a = \Phi \text{Ad}_{U_a}$,

$$\Phi_{a_t} \cdots \Phi_{a_1} = \Phi^t \text{Ad}_{U_{a_1 + \cdots + a_t}}. \quad (7)$$

Thus equal-length words with equal displacement have equal nonselective evolution. Different lengths retain different powers of Φ , unless it is predictively invisible or idempotent. A cancelling detour may return to the same base point but alter the outcome-discarded ensemble. This is a coarse-grained coordinate, not part of the fully observed conditional state.

For a finite separating battery, define

$$\epsilon_{\text{path}}(L) = \max_{\substack{|w|, |w'| \leq L \\ \Delta(w) = \Delta(w')}} \max_{h, \tau} |p(\tau \mid h, w) - p(\tau \mid h, w')|.$$

Zero is path closure on the tested predictive quotient. Equal-length and all-length residuals must be reported separately.

5 The Hesse SIC and Weyl translations

5.1 Nine nonorthogonal landmarks in a qutrit

Let $\omega = e^{2\pi i/3}$, with

$$X|j\rangle = |j+1 \bmod 3\rangle, \quad Z|j\rangle = \omega^j |j\rangle.$$

They obey $ZX = \omega XZ$. Although the vector representation is projective, global phases disappear under conjugation, so Ad_X and Ad_Z commute.

From

$$|\psi_{00}\rangle = \frac{(0, 1, -1)^T}{\sqrt{2}},$$

define

$$|\psi_{mn}\rangle = X^m Z^n |\psi_{00}\rangle, \quad \Pi_{mn} = |\psi_{mn}\rangle\langle\psi_{mn}|.$$

Direct calculation yields the Hesse SIC identities

$$\frac{1}{3} \sum_s \Pi_s = I, \quad \text{Tr}(\Pi_s \Pi_t) = \begin{cases} 1, & s = t, \\ 1/4, & s \neq t. \end{cases} \quad (8)$$

Thus a qutrit supports nine operational landmarks without nine mutually orthogonal basis states [3].

Let

$$\mathcal{A} = \{I, X, X^\dagger, Z, Z^\dagger\}.$$

Conjugation permutes SIC projectors:

$$U_a \Pi_s U_a^\dagger = \Pi_{s+a}.$$

The four nontrivial actions are cardinal generators of the 3×3 Cayley torus.

5.2 Integrated Hesse measure-and-prepare branches

Define

$$K_o^{(a)} = \frac{1}{\sqrt{3}} \Pi_o U_a, \quad \mathcal{E}_o^a(\rho) = K_o^{(a)} \rho K_o^{(a)\dagger}. \quad (9)$$

Completeness follows from Eq. (8):

$$\sum_o K_o^{(a)\dagger} K_o^{(a)} = U_a^\dagger \left(\frac{1}{3} \sum_o \Pi_o \right) U_a = I.$$

For input Π_s ,

$$p_a(o | s) = \begin{cases} 1/3, & o = s + a, \\ 1/12, & o \neq s + a, \end{cases} \quad \rho_{a,o} = \Pi_o. \quad (10)$$

The likelihood peak translates, while the outcome becomes the next state. The last token is an exact predictive sufficient statistic.

For a uniform prior, the outcome marginal is uniform and immediate information is

$$I_{\text{Hesse}} = \log_2 9 - H \left(\frac{1}{3}, \underbrace{\frac{1}{12}, \dots, \frac{1}{12}}_8 \right) = 0.2516291674 \text{ bits.}$$

6 Terminal semantics and exact Bellman geometry

6.1 Two different goals

Reinforcement learning assigns meaning to states only relative to rewards, costs, and termination [8, 9]. Here each nonterminal intervention costs one. The complete action, initial-state, and goal specification is in Section 2.2. Two superficially similar tasks differ:

Report again. Even at g , the agent must act and obtain outcome g . This asks for external recertification and gives shells $(4, 4, 5)$.

State hitting. The episode ends when the current predictive state is g , so $V_g(g) = 0$. This asks the agent to occupy the goal state.

The earlier self-edge no-go was correct for report-again semantics, but it is not a no-go for state-hitting hodology.

6.2 Exact Hesse hitting values

Theorem 6.1 (State-hitting Hesse shells). *For the instrument Eq. (9), unit action cost, and the boundary $V_g(g) = 0$, the optimal values are*

$$\boxed{v_0 = 0, \quad v_1 = 4, \quad v_2 = 5,}$$

where shells are torus word distances zero, one, and two from g .

Proof. Symmetry reduces nine states to three shells. The goal boundary gives $v_0 = 0$. From an edge, move its likelihood peak to g . The success probability is $1/3$. The four edge and four diagonal failure outcomes each have probability $1/12$, so

$$v_1 = 1 + \frac{1}{3}v_1 + \frac{1}{3}v_2. \quad (5.1)$$

From a diagonal state, a cardinal move peaks at an edge. The off-peak goal outcome has probability $1/12$. The peak and three other edge outcomes contribute $1/3 + 3/12 = 7/12$; four diagonal outcomes contribute $1/3$. Thus

$$v_2 = 1 + \frac{7}{12}v_1 + \frac{1}{3}v_2. \quad (5.2)$$

Solving gives $v_1 = 4, v_2 = 5$. Direct comparison of the five action rows verifies the minimizing actions. $\square$

Corollary 6.2. *The directed hitting cost $D(s, g) = V_g(s)$ is a translation-invariant metric on $\mathbb{Z}_3^2$.*

Proof. Translation covariance and inverse controls give symmetry. Off-diagonal values are positive. The only nontrivial triangle inequality is $5 \leq 4 + 4$. $\square$

6.3 A general metric theorem

Theorem 6.3 (Covariant hitting costs induce a group metric). *Consider a controlled Markov process on a finite group G . Assume:*

1. *transition laws are translation covariant;*
2. *every nonterminal action costs at least $c_{\min} > 0$;*
3. *each goal can be reached under a proper policy;*
4. *every policy has an equal-cost inverse-reflected policy;*
5. *$V_g(g) = 0$ is an absorbing hitting boundary.*

Then $D(s, g) = V_g(s)$ is a translation-invariant metric on G .

Proof. Nonnegativity is immediate. If $s \neq g$, any successful trajectory pays at least $c_{\min}$, establishing strictness. Translation covariance gives $V_g(s) = v(g - s)$, while inverse reflection gives symmetry. Concatenate an ε -optimal policy from s to u with one from u to g . The strong Markov property at the hitting time of u gives

$$V_g(s) \leq V_u(s) + V_g(u) + 2\varepsilon.$$

Let $\varepsilon \downarrow 0$ to obtain the triangle inequality. $\square$

Remark 6.4. This proves metricity, not two-dimensional Euclidean embeddability. It also depends essentially on state hitting. Under report-again semantics, if an edge's aligning action has the same complete outcome/posterior kernel and cost as an action at the goal, their Bellman right-hand sides coincide and the self-edge collapse returns, regardless of branch Choi rank.

7 Global torus metric versus a local plane

7.1 Schoenberg's exact Euclidean criterion

For a finite symmetric distance matrix D , define

$$J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T, \quad B = -\frac{1}{2}J(D^{\circ 2})J.$$

Schoenberg's theorem states that D is a Euclidean distance matrix in $\mathbb{R}^k$ exactly when $B \succeq 0$ and $\text{rank } B \leq k$ [10]. If $B = Q\Lambda Q^T \succeq 0$, coordinates are $Q\Lambda^{1/2}$. Negative eigenvalues disprove exact Euclidean embeddability; retaining two positive eigenvalues gives classical multidimensional scaling (MDS), not an exact proof [11].

Proposition 7.1 (Schoenberg spectrum of the Hesse hitting metric). *The distance 0, 4, 5 is Euclidean, but its minimal embedding dimension is eight. Its centered Gram eigenvalues are*

$$0, \quad \underbrace{17, \dots, 17}_4, \quad \underbrace{\frac{7}{2}, \dots, \frac{7}{2}}_4.$$

Proof. D^2 is 16 on four cardinal displacements and 25 on four diagonals. For a character $(k, l) \in \mathbb{Z}_3^2$, define $A_0 = 2$, $A_1 = A_2 = -1$. The Fourier eigenvalue of D^2 is

$$\widehat{D^2}(k, l) = 16(A_k + A_l) + 25A_kA_l.$$

For nontrivial characters with exactly one zero coordinate this is -34 ; when both coordinates are nonzero it is -7 . On the centered subspace, B has minus one half these eigenvalues. All eight are positive. $\square$

The group and neighbor graph are two-dimensional, but the global intrinsic metric is not a planar 3×3 patch. Periodic boundary points are neighbors. This is a topology/embedding distinction, not a failure of the torus interpretation.

7.2 Local flatness

For a larger $m \times m$ torus, graph balls of radius $r < m/2$ lift uniquely to the universal cover before wraparound. Local flatness should be tested on these patches. Even then, cardinal step costs produce a Manhattan norm, which is flat Finsler rather than Euclidean. Euclidean local geometry requires diagonal costs $\sqrt{2}$ times edge costs on elementary squares and compatible higher-point constraints.

8 Two distinct retained-memory mechanisms

8.1 Observed memory/reset family: finite and exactly soluble

The selectable buttons and ten observed branch tokens are summarized in [Section 2.2](#); in particular, the agent selects (a), not the branch below.

The principal search family has ten observed outcomes: one token r means that the memory branch occurred, and nine tokens (q, o) mean reset to Hesse state o . For action a ,

$$K_{a,r} = \sqrt{\mu} U_a, \quad 0 \leq \mu \leq 1. \quad (11)$$

This Kraus operator has full matrix rank and translates $\Pi_s \mapsto \Pi_{s+a}$ exactly.

Reset outcomes use

$$F_o^{(\xi)} = \xi \frac{\Pi_o}{3} + (1 - \xi) \frac{I}{9}, \quad 0 \leq \xi \leq 1,$$

and the CP branch

$$\mathcal{E}_o^a(\rho) = (1 - \mu) \text{Tr}[F_o^{(\xi)} U_a \rho U_a^\dagger] \Pi_o. \quad (12)$$

A Kraus realization is

$$K_{a,o,j} = \sqrt{1 - \mu} |\psi_o\rangle \langle j| \sqrt{F_o^{(\xi)}} U_a, \quad j = 0, 1, 2.$$

Since $\sum_o F_o^{(\xi)} = I$, the memory and reset effects sum to $\mu I + (1 - \mu)I = I$. For $\xi < 1$, each reset map has Choi rank three.

Every observed branch returns a Hesse state: r gives $s + a$, while reset outcome o gives o . Starting from a known Hesse token, the observed history determines the current predictive class exactly. Memory is physical rather than an external counter, because the translated input survives in the qutrit on branch r .

The reset-mode choice is state independent. At $\xi = 1$,

$$I(S; O | A) = (1 - \mu) I_{\text{Hesse}}.$$

At general ξ ,

$$P_a^{(\xi)}(o | s) = \xi P_a^{\text{Hesse}}(o | s) + \frac{1 - \xi}{9},$$

and its entropy gives the exact immediate information.

8.2 Exact Bellman equations

Translation symmetry again leaves edge E and diagonal D . Put

$$c = \frac{(1 - \mu)(4 - \xi)}{9}, \quad h = \mu + \frac{(1 - \mu)(16 + 5\xi)}{36}, \quad A = 1 - c.$$

Moving an edge toward the goal and a diagonal toward an edge yields

$$E = 1 + c(E + D), \quad D = 1 + hE + cD. \quad (13)$$

Solving,

$$\boxed{E = \frac{1}{A^2 - ch}}, \quad \boxed{D = \frac{1 + hE}{A}}. \quad (14)$$

Full nine-state Bellman iteration agrees with these expressions to numerical precision throughout the search.

For sharp reports, $\xi = 1$, the result simplifies:

$$\boxed{E = \frac{4}{(1 + \mu)^2}}, \quad \boxed{D = \frac{3\mu + 5}{(1 + \mu)^2}}. \quad (15)$$

Consequently

$$D - E = \frac{3\mu + 1}{(1 + \mu)^2} > 0, \quad D \leq 2E \quad (0 \leq \mu \leq 1).$$

Identity of indiscernibles, strict shell ordering, and the only nontrivial triangle inequality therefore hold analytically across the family.

The endpoints isolate the tradeoff:

$$\mu = 0 : \quad (E, D) = (4, 5), \quad I = I_{\text{Hesse}}; \quad \mu = 1 : \quad (E, D) = (1, 2), \quad I = 0.$$

8.3 An exact elementary Euclidean square

Four states generated by two independent cardinal steps form a Euclidean square exactly when $D/E = \sqrt{2}$. From Eq. (15),

$$\frac{D}{E} = \frac{3\mu + 5}{4}.$$

Therefore

$$\boxed{\mu_{\square} = \frac{4\sqrt{2} - 5}{3} = 0.2189514165 \dots} \quad (16)$$

gives an exact local square. Its immediate information is

$$(1 - \mu_{\square})I_{\text{Hesse}} = 0.1965346 \text{ bits}.$$

This exact four-point statement does not unwrap the global nine-state torus.

8.4 Two-Kraus retained-memory interpolation

A second analytic family, useful for future belief-state optimization, is

$$\mathcal{J}_o^{\lambda}(\rho) = \frac{\lambda}{3}\Pi_o\rho\Pi_o + \frac{1 - \lambda}{9}\rho, \quad (17)$$

applied after the control U_a . Its two Kraus operators are proportional to $\Pi_o U_a$ and U_a , so each branch has Choi rank two when $0 < \lambda < 1$. The posterior is proportional to

$$\frac{\lambda}{3}\Pi_o U_a \rho U_a^{\dagger} \Pi_o + \frac{1 - \lambda}{9} U_a \rho U_a^{\dagger}.$$

Unlike Eq. (12), the same outcome retains a continuous source-dependent component. On Hesse inputs,

$$p_{\lambda}(o \mid s, a) = \begin{cases} (1 + 2\lambda)/9, & o = s + a, \\ (4 - \lambda)/36, & o \neq s + a. \end{cases}$$

This formula has been audited at both endpoints: it is uniform at $\lambda = 0$ and gives $1/3, 1/12$ at $\lambda = 1$.

The family is exact and covariant, but its posterior generally leaves the nine-state orbit. It requires density-state or PSR Bellman planning. It must not be confused with the observed memory/reset family, whose observed branches close exactly on nine Hesse states.

8.5 Lüders-like latent-memory family used in learning

The opaque learning experiment instead uses effects

$$E_o^\eta = \eta \frac{\Pi_o}{3} + (1 - \eta) \frac{I}{9}$$

and one full-matrix-rank Lüders Kraus operator

$$K_{a,o}^\eta = \sqrt{E_o^\eta} U_a.$$

For $0 < \eta < 1$, the posterior conditioned on o depends on the incoming state. The branch is not an outcome-only preparation, and the last outcome need not be sufficient. This construction is sometimes informally called higher rank because the effect and Kraus matrix have support beyond one ray; its CP map has a single Kraus operator. That convention is distinct from the strict Choi-rank-two meaning in [Section 8.4](#).

9 Deterministic search over the exact finite family

9.1 Search protocol and selection

A deterministic 20×19 grid scanned 380 pairs (μ, ξ) . Every pair was evaluated for Kraus completeness, covariance, immediate mutual information, analytic and iterative Bellman agreement, metric axioms, scaled torus distortion, Schoenberg spectrum, MDS stress, and path closure. The deterministic stated high-memory selection rule required

$$\mu > 0, \quad I(S; O | A) \geq 0.05 \text{ bits},$$

a valid metric, and minimal relative RMSE after scaling the torus word metric. It selected

$$\boxed{(\mu, \xi) = (0.8, 1)}.$$

Table 1: Exact and diagnostic results for principal finite-family candidates. RMSE is after the best common scale to torus word distance.

Candidate	μ	ξ	MI (bits)	E	D
Hesse hitting baseline	0	1	0.251629	4.000000	5.000000
Exact local square	0.218951	1	0.196535	2.692075	3.807169
Selected torus	0.8	1	0.050326	1.234568	2.283951
Null report	0.8	0	0	1.331361	2.396450
Weak report	0.8	0.5	0.014477	1.281595	2.338911
Memory only	1	1	0	1.000000	2.000000

The selected candidate has strict shell margin $D - E = 1.049383$, scaled torus relative RMSE 0.033287, and held-out diagonal nonadditivity $(2E - D)/D = 0.0811$. Correlation with torus graph distance is exactly one whenever $0 < E < D$, because there are only two nonzero shells; it is therefore uninformative by itself. Distortion and additivity expose the remaining difference.

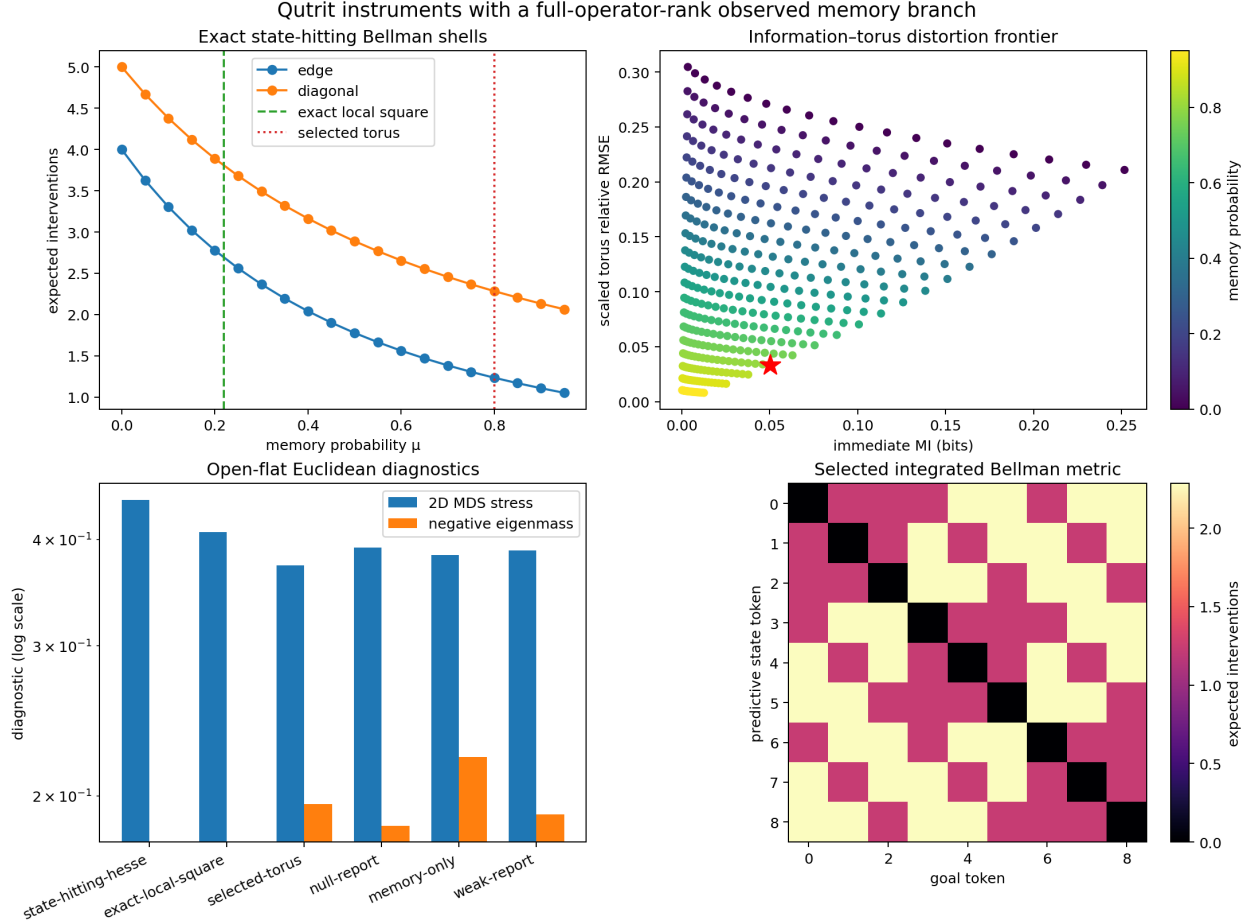


Figure 1: Exact covariant-memory search. Top left: analytic Bellman shells across the sharp family, including the exact-square and selected values. Top right: immediate information versus scaled torus distortion over all 380 candidates. Bottom left: global embedding diagnostics prevent local-square claims from being extended to a planar torus. Bottom right: the selected all-pairs hitting metric.

9.2 Embedding and closure diagnostics

The selected torus has negative Schoenberg eigenmass fraction 0.1957, four positive centered dimensions, and two-dimensional MDS stress 0.3731. The exact local-square cell has zero negative eigenmass and rank two, while its full nine-token matrix has stress 0.4082. Correlation with an open 3×3 grid is only 0.5276, as expected from periodic boundaries.

All equal-length displacement classes through word length four close at floating-point precision. The selected all-length trace-distance residual is 0.31866. The Hesse reset endpoint is even more path-age sensitive (0.66406); memory-only motion closes at numerical precision. These numbers show length dependence of the outcome-discarded ensemble channel. They do not establish a noise-age coordinate for the fully observing agent, whose conditional branches close on pure Hesse rays.

9.3 Observable action learning and labelled string validation

The primary action-learning protocol supplies no successor-state labels. Each trial first uses a sharp Hesse anchor instrument, whose opaque token physically prepares the corresponding ray. The agent applies an opaque action and retains trials on which the observed memory branch occurs. It then applies the same sharp Hesse instrument as a common future probe. The learner’s entire record is therefore

(opaque anchor token, opaque action token, opaque future-report token).

At $(\mu, \xi) = (0.8, 1)$, 17,982 of 22,500 attempts had the memory outcome. Maximum-likelihood response maps are all bijections, commute, generate a group and orbit of order nine, and agree with the hidden physical maps with 100% accuracy when scored only after learning. The anchor and probe define an operational frame; their arbitrary token permutation remains gauge.

For comparison, a separate labelled benchmark tests finite-state estimation:

With oracle-labelled Hesse source and successor classes, a transition table trained on 400 samples per pair was evaluated on 2,000 independent length-eight strings:

exact filter NLL	1.3132 bits/event,
learned kernel NLL	1.3989 bits/event,
state-marginal NLL	3.8964 bits/event.

The 0.0857-bit oracle gap measures finite training error. The large advantage over the marginal model shows that learned state and action meaning generalize beyond one-step fitting.

10 Opaque predictive learning

10.1 Experimental design

The initialization and downstream goal convention are stated explicitly in [Section 2.5](#).

The second experiment receives only shuffled action IDs and shuffled outcome tokens. It receives no coordinates, density matrices, Kraus operators, or displacement names. For each of six models, 12,000 training and 3,000 held-out seven-step sequences were generated with seed 20260812.

Three transparent predictors were fitted:

1. an action-only marginal model;
2. last outcome plus next action;
3. previous action/outcome, current outcome, and next action.

The exact density-matrix filter is an oracle, not an input to learning. Transition kernels were estimated from adjacent tokens; maximum-probability maps were tested for bijectivity, orders, commutation, group size, and orbit. Invalid nonbijective maps were assigned group order zero rather than being misreported as a transformation group.

10.2 Topology and Bellman recovery

The rank-one baseline recovers five bijections: one identity and four commuting order-three maps. They generate a transitive group of order nine. Its learned state-hitting shells are approximately

$$(0, 3.9553, 4.9232),$$

near the exact $(0, 4, 5)$. Both Weyl-covariant Lüders models also recover a bijective, commuting order-nine action group. Token-aggregated Bellman shells are strictly ordered:

$$\eta = 0.55 : (0, 6.6285, 7.0926), \quad \eta = 0.80 : (0, 5.1137, 5.9145).$$

For these latent-memory models, token Bellman planning is an approximation because the last token is not a sufficient causal state.

Table 2: Held-out prediction and learned topology. NLL is bits per outcome. The last column is the exact quantum-filter oracle.

Model	Last token	Two-event	Oracle	Group	Orbit
Rank-one reset	2.9568	2.9638	2.9549	9	9
Weyl Lüders $\eta = .55$	3.1491	3.1568	3.1137	9	9
Weyl Lüders $\eta = .80$	3.0740	3.0793	3.0172	9	9
Null Lüders	3.1734	3.1796	3.1699	0	1
Haar Lüders $\eta = .55$	3.1495	3.1581	3.1164	0	1
External-DFA null	3.1729	3.1803	3.1699	0	1

10.3 What prediction reveals about memory

At $\eta = 0.55$, the last-token/oracle gap is 0.03535 bits per outcome; at $\eta = 0.80$, it is 0.05683. The exact quantum filter uses information that the last token discards. This is operational evidence of latent memory.

The naive two-event estimator did not exploit it: its held-out NLL was slightly worse than last-token prediction. Raw future-law variation among histories sharing a last token was comparable to the null sampling floor; null-subtracted means were 0.00093 at $\eta = .55$ and -0.00134 at $\eta = .80$. It would be incorrect to call this noisy suffix statistic a learned fiber. The oracle gap establishes that memory exists; learning it requires a regularized PSR rather than context proliferation.

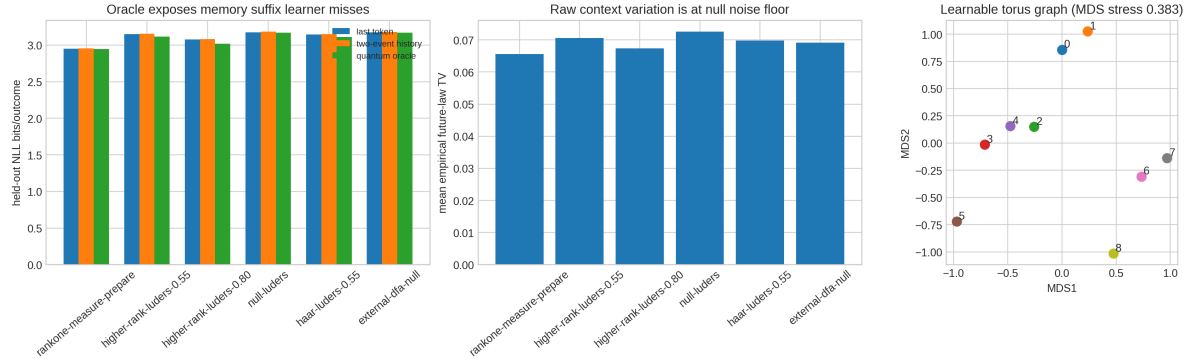


Figure 2: Opaque learning diagnostics. Left: exact quantum filtering exposes predictive memory that a finite suffix model misses. Center: raw context variation is at the null noise floor and is not claimed as learned memory. Right: the learned torus graph is topologically correct, but its planar MDS display has nonzero stress.

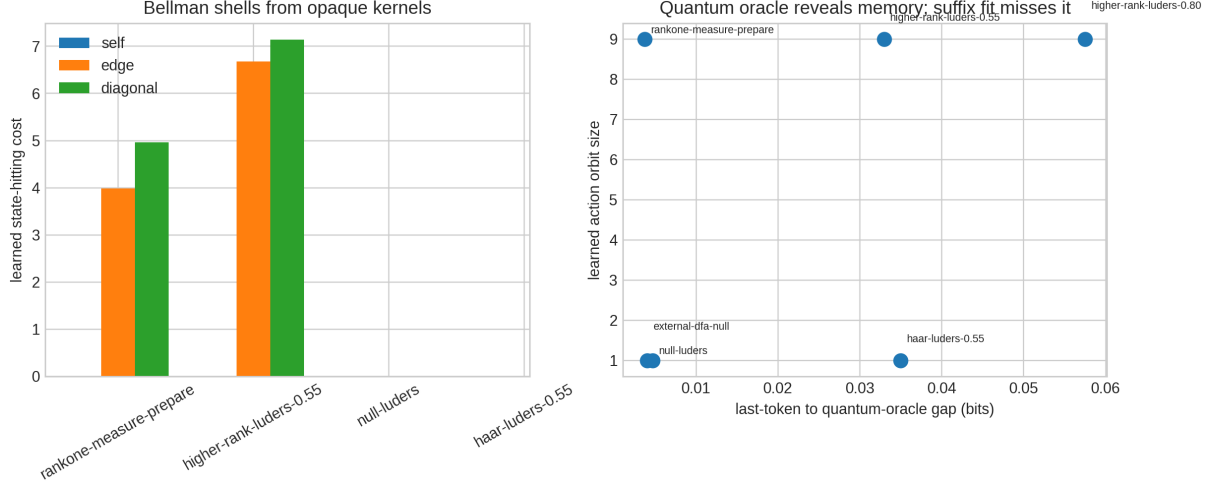


Figure 3: Left: strict state-hitting shells recovered from opaque token kernels. Right: spatial action topology and longer-history prediction are independent tests. Null, Haar, and external-DFA controls fail bijective translation-group recovery.

10.4 Matched controls

Rank-one Hesse reset. Positive topology baseline with an exact last-token predictive quotient but no retained branch.

Null Lüders. Uniform outcomes carry no state information and inferred argmax maps are nonbijective.

Haar Lüders. Actions can affect future predictions but do not close into the required commuting translation group.

External DFA. A genuine nine-node classical register undergoes exact $\mathbb{Z}_3^2$ transitions while quantum tokens remain i.i.d. uniform. The token-only interface rejects geometry; explicitly exposing the register recovers its hand-coded group and records its classical provenance.

Memory only. The exact word metric 0, 1, 2 and perfect path closure coexist with zero immediate information.

Permutation gauge. Persistent shuffling changes names, not group order, commutation, orbit, or Bellman shells. Episode-wise reshuffling would destroy persistent meaning.

11 Information, disturbance, and geometry

11.1 Immediate information is not operational action meaning

For a state label S , outcome O , and fixed action a ,

$$I(S; O \mid A = a) = \sum_{s,o} \pi_s p(o \mid s, a) \log_2 \frac{p(o \mid s, a)}{\sum_t \pi_t p(o \mid t, a)}.$$

This measures the current report. Future-operational action meaning instead asks whether later test probabilities change. The memory-only endpoint has zero immediate information but perfect future translation meaning. The null instrument has neither. The Haar control has future meaning but the wrong algebra. These examples separate three properties that a single mutual information number cannot summarize.

Channel distinguishability may be quantified by the diamond norm $\|\Phi_a - \Phi_b\|_\diamond$, while future outcome distributions may use total variation or Jensen–Shannon divergence [12]. These measure distinguishability, not spatial covariance.

11.2 No exact universal information without disturbance

Proposition 11.1. *If an instrument’s nonselective channel equals a unitary channel $\mathcal{U}(\rho) = U\rho U^\dagger$ on the full state space, then each branch is $\mathcal{E}_o = p_o\mathcal{U}$. Outcomes are input independent.*

Proof. The Choi matrix of a unitary channel has rank one. Positive branch Choi matrices sum to it, so every branch has support in that same one-dimensional space: $J(\mathcal{E}_o) = p_o J(\mathcal{U})$. The Choi isomorphism gives the result. $\square$

The assumptions are exact and universal. Restricted commuting state families, encoded subsystems, or approximate preservation allow information–disturbance tradeoffs [13, 14]. In this project, increasing reset probability increases immediate information but also produces more path-dependent noise.

11.3 A base/fiber interpretation

The learned permutations supply a base projection

$$\pi : \mathcal{S}_{\text{predictive}} \longrightarrow \mathbb{Z}_m^2.$$

Predictive states with the same base token may differ in conditional density, purity, path length, or uncertainty. Those residual degrees form a fiber $\mathcal{F}_x = \pi^{-1}(x)$. An action ideally has:

- a horizontal component translating x ;
- a vertical/internal component evolving fiber state;
- a coupling that may depend covariantly on the base.

Equation Eq. (7) makes a coarse-grained version concrete. Net displacement determines the base, while Φ^t records noise age only for an observer who discards branch outcomes. A genuine predictive fiber appears instead in the Lüders family, where histories with the same last token can have different future laws; the oracle detects it, but the suffix learner does not yet reconstruct it.

Operational holonomy can be studied from learned PSR action operators B_a . The loop

$$\mathcal{K}_{ab} = B_a B_b B_a^{-1} B_b^{-1}$$

transforms by similarity under PSR gauge. Its spectrum and future-test effect are gauge invariant. Trivial base commutators with nontrivial fiber action would provide a first operational analogue of a connection.

12 What has and has not been established

12.1 Exact positive statements

1. Correct state-hitting semantics converts the integrated Hesse result from nonmetric (4, 4, 5) to metric (0, 4, 5).
2. The Hesse metric recovers a homogeneous $\mathbb{Z}_3^2$ torus neighbor graph.
3. Its global Euclidean embedding dimension is exactly eight.
4. The observed memory/reset family is CPTP, covariant, informative for $\mu < 1, \xi > 0$, and exactly closed on nine predictive Hesse states.

5. Its Bellman shells have the closed forms Eqs. (14) and (15); strictness and triangles hold for the entire sharp family.
6. The value $\mu_{\square}$ gives exact Euclidean elementary squares.
7. Equal-length word closure follows analytically from covariance.

12.2 Numerical statements

1. The constrained grid search selects a positive-information candidate with 3.33% scaled torus distortion.
2. An observable anchor/action/future-probe protocol recovers the exact family’s five maps and order-nine group without successor-state labels. Oracle-labelled trials separately validate held-out length-eight tables.
3. Opaque Lüders experiments recover a commuting order-nine group and strict token-level hitting shells.
4. Exact quantum-filter likelihood exposes latent memory not captured by the last token.
5. Null, Haar, and external-memory controls fail at their predicted operational layer.

12.3 Limitations

1. Equal displacement does not imply equal predictive state across different path lengths. The selected all-length residual is 0.319.
2. A 3×3 torus is not an open planar 3×3 grid. Only the elementary four-state square is exactly planar in the present construction.
3. The observed memory/reset outcome grammar is engineered. It has not emerged from unconstrained instrument optimization.
4. The suffix learner does not recover the latent fiber. The oracle demonstrates its existence, not its learned representation.
5. Token-aggregated Bellman values for Lüders models are approximations to belief-state values.
6. Immediate information, metric quality, and memory retention are multiobjective; no canonical scalar weighting has been justified.
7. Finite torus graph distance is locally Manhattan. Calling it Riemannian would overstate the result.

13 Future research

13.1 Larger phase tori and honest flat patches

A qutrit supports arbitrarily many nonorthogonal phase landmarks:

$$|\phi_{xy}^{(m)}\rangle = \frac{|0\rangle + \zeta^x|1\rangle + \zeta^y|2\rangle}{\sqrt{3}}, \quad \zeta = e^{2\pi i/m},$$

with commuting controls

$$U_x = \text{diag}(1, \zeta, 1), \quad U_y = \text{diag}(1, 1, \zeta).$$

For $m = 5, 7, 9$, nonwrapping patches of radius $r < m/2$ contain multiple plaquettes. A covariant phase POVM and positive Choi seed can then be optimized for information, strict hitting costs, local isotropy, and weak fiber disturbance.

13.2 Spectral PSRs

Suffix tables proliferate contexts and performed worse at the available sample size. The next learner should:

1. collect controlled history/test Hankel blocks over increasing horizons;
2. select rank with bootstrap singular-value intervals and held-out shrinkage;
3. fit low-rank action/outcome update operators;
4. test shifted-Hankel action separation and group relations in PSR gauge;
5. solve goal problems in learned predictive state rather than last-token state;
6. compare with the exact quantum filter only as a calibration oracle.

Spectral PSRs expose rank and linear gauge explicitly [4, 5]. Recurrent models can follow after the transparent baseline.

13.3 Instrument optimization

Use [Theorem 4.1](#) with $C_0 = LL^\dagger$, enforcing covariance and CPTP exactly. Optimize a Pareto frontier rather than one opaque loss:

$$\max\{I_{\text{immediate}}, I_{\text{future}}, M, \Delta_{\text{edge}}\}, \quad \min\{1 - F_{\text{move}}, \epsilon_{\text{path}}, \epsilon_{\text{flat}}\}.$$

Here M is same-outcome future separation, F_{move} is fidelity to the intended unitary move, and ϵ_{flat} measures translated patch inconsistency. Search should begin from the observed and two-Kraus analytic families before allowing an unrestricted Choi seed.

13.4 Toward bundles and curvature

The spatial base should be the coarsest predictive quotient preserving place-like goal values and local action increments. Posterior uncertainty, measurement context, goal progress, and quantum memory remain in the fiber. Loop transformations can probe connection and curvature. Eventually, state-dependent local cost tensors and nontrivial holonomy could connect hodological geometry to dynamical spacetime models. At the present stage, these are research directions, not conclusions.

14 Conclusion

A qutrit is large enough to support an operationally learned two-dimensional translation topology and a strict goal metric without nine orthogonal basis states. The key correction is conceptual: being at a goal and being required to report it again are different reinforcement-learning tasks. Standard state-hitting semantics gives the exact integrated Hesse metric (0, 4, 5).

Retained memory then supplies a controlled interpolation. The observed memory/reset family joins positive immediate information, exact finite conditional states, covariant translations, and analytic Bellman geometry. One parameter value gives exact local squares; another gives a high-memory torus approximation selected by a transparent rule. A common future probe recovers its action group from opaque observed tokens, and a separate opaque Lüders experiment recovers the same torus topology from histories.

The Lüders oracle gap identifies latent predictive memory over the spatial token base, but the present learner does not yet reconstruct that candidate fiber. The next decisive step is to learn it with a spectral PSR on a larger torus and test translated nonwrapping patches. Geometry should be claimed only after prediction, action algebra, Bellman metricity, topology, and local embedding all pass separate controls.

A Reproducibility and acceptance criteria

A.1 Commands and artifacts

From the search directory, run the seven focused unit tests and deterministic search:

```
python -m unittest discover -s tests -v
MPLBACKEND=Agg python run_search.py
```

From the learning directory:

```
python -m unittest -v test_opaque_learning.py
MPLBACKEND=Agg python opaque_learning.py --train 12000 \
--test 3000 --seed 20260812
```

The search archive contains all 380 grid points, candidate diagnostics, path closure tables, all-pairs Bellman matrices, a JSON manifest, and [Fig. 1](#). The learning archive contains summary and action-algebra tables, per-model transition kernels, Hankel singular spectra, a manifest, and [Figs. 2 and 3](#).

A.2 Acceptance table

Table 3: Preregistered-style criteria for a future covariant-memory geometry.

Layer	Statistic	Acceptance
Physical	Kraus completeness and covariance residual	Below 10^{-10}
Prediction	Held-out NLL and calibration	Better than matched marginal/null
Memory	Same-outcome future separation	Bootstrap interval above zero
Action	Shifted-Hankel pair separation	Every intended distinct pair
Algebra	Orders, commutators, orbit	$\mathbb{Z}_m^2$ within uncertainty
Hitting	Bellman solution	Finite; residual below 10^{-10}
Metric	Strictness, symmetry, triangles	No significant violations
Topology	Lowest-cost neighbor graph	Torus isomorphism
Local geometry	Patch residual and increment variance	Stable below declared threshold
Controls	Null, cloned, Haar, DFA tests	Fail at predicted layer

B Derivation audit for the sharp family

Substituting $\xi = 1$ in [Eq. \(13\)](#) gives

$$c = \frac{1 - \mu}{3}, \quad h = \mu + \frac{7(1 - \mu)}{12} = \frac{7 + 5\mu}{12}, \quad A = \frac{2 + \mu}{3}.$$

Then

$$A^2 - ch = \frac{(2 + \mu)^2}{9} - \frac{(1 - \mu)(7 + 5\mu)}{36} = \frac{(1 + \mu)^2}{4},$$

which yields $E = 4/(1 + \mu)^2$. Further,

$$D = \frac{1 + hE}{A} = \frac{3\mu + 5}{(1 + \mu)^2}.$$

At $\mu = 0$, these reproduce [Theorem 6.1](#); at $\mu = 1$, deterministic memory motion yields one cardinal step or two.

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